

A FANO PLANE IN THE SPECIAL FIBRE OF AN EXPLICIT M_{23} -COVER

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ABSTRACT. Huang, Jackson, Lee, Poonen, Pries and Zhang recently completed the realization of all sporadic simple groups as Galois groups over $\mathbf{Q}$ by constructing an explicit M_{23} -extension of $\mathbf{Q}(t)$. Their construction uses a seven-element Nielsen class on which the absolute Galois group has a fixed point. Here a Nielsen class is a set of generating branch-cycle triples with prescribed conjugacy classes, modulo simultaneous conjugation. We do not explain or reprove that fixed point. Instead, we determine arithmetic structure in the fibre over the rational branch point $T = 0$ of their published cover.

For the integral plane equation $F_{\text{int}}(T, V)$, the special fibre factors as

$$F_{\text{int}}(0, V) = 23^4 H_7(V) R_8(V)^2,$$

where H_7 and R_8 are irreducible of degrees 7 and 8. The roots of H_7 are the seven unramified points over $T = 0$. Their Galois group is $\text{PSL}_3(2)$. Each of its 21 involutions fixes three roots; there are seven distinct fixed triples, each occurring for three involutions, and these triples are the lines of a Fano plane. A degree-seven resolvent gives the corresponding line field. The point and line fields are nonisomorphic but arithmetically equivalent, meaning that they have the same Dedekind zeta function.

The roots of R_8 are the eight ramification points, represented in the specialized equation by double roots. Starting from the splitting field of H_7 , adjoining these eight roots gives an extension with Galois group $2^3:\text{PSL}_3(2)$. Separating the two local branches at the eight ramification points gives one further quadratic extension. Altogether we write $E = \text{Spl}_{\mathbf{Q}}(H_7)$ and $L_0 = \text{Spl}_{\mathbf{Q}}(H_7 R_8)$ and obtain a tower

$$\mathbf{Q} \subset E \subset L_0 \subset M, \quad \text{Gal}(M/\mathbf{Q}) \cong C = C_{M_{23}}(x) \cong 2^4:\text{PSL}_3(2),$$

with successive relative Galois groups $\text{PSL}_3(2)$, $(\mathbf{Z}/2)^3$, and $\mathbf{Z}/2$. The field $L_0 = M^{(x)}$ is the splitting field of $H_7 R_8$, and the last quadratic extension records which of the two local branches is chosen. These extensions are unramified outside $\{2, 3, 23\}$.

The eight roots of R_8 form an affine torsor under $(\mathbf{Z}/2)^3$, meaning that this group acts freely and transitively on them. Translation classes of ordered affine frames form a canonical set of size 168, with a free transitive $\text{PSL}_3(2)$ -action. We also record the generic–special-fibre comparison: tame specialization identifies the Galois action on the roots of R_8 with the action induced on the eight two-sheet inertia orbits of the generic fibre. A companion paper uses this affine-frame torsor to construct an outer-twisted point–line correlation. The comparison here uses the known rational model and does not give an independent proof of the Galois fixed point.

An involution of $\text{PSL}_3(2)$ determines a flag, meaning an incident point–line pair, of the Fano plane. We use this observation to describe complex conjugation and suitable Frobenius elements. Finally, we return to the original seven Nielsen classes: complex conjugation fixes three of them, and for each fixed class we compute the unique element of C that compares the class with its complex conjugate.

The finite-group and number-field assertions are accompanied by exact GAP and PARI/GP certificates. The calculation of the ordinary nodes of the plane model is cited from a separate source.

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1. INTRODUCTION

1.1. The fixed-point phenomenon. In [1] the authors complete a program begun in the 1980s by proving that the Mathieu group M_{23} , the last of the twenty-six sporadic simple groups to resist, occurs as a Galois group over $\mathbf{Q}$. Their method is to exhibit an explicit regular M_{23} -extension of $\mathbf{Q}(t)$, obtained from a numerically computed Belyi map.

The construction uses a *non-rigid* triple of conjugacy classes. Concretely, let

$$\mathbf{c} = (2A, 23A, 23B)$$

and let $\text{Ni}_{\mathbf{c}}$ be the set of generating triples (g_1, g_2, g_3) of M_{23} with g_i in the i -th class and $g_1 g_2 g_3 = 1$, taken modulo simultaneous conjugation. Then $|\text{Ni}_{\mathbf{c}}| = 7$: by the Riemann existence theorem there are exactly seven M_{23} -covers of $\mathbf{P}_{\mathbf{C}}^1$ branched over a fixed three-element set with this local monodromy. Rigidity would ask for $|\text{Ni}_{\mathbf{c}}| = 1$; it fails.

The absolute Galois group $\mathfrak{S}_{\mathbf{Q}}$ permutes these seven covers. As [1, §1.4] puts it:

“One might expect $\mathfrak{S}_{\mathbf{Q}}$ to act transitively on these seven. Miraculously, instead $\mathfrak{S}_{\mathbf{Q}}$ has a fixed point! We do not have a conceptual explanation for this.”

The fixed point has field of moduli $\mathbf{Q}$, descends, and yields the desired extension. The results below describe arithmetic attached to the published cover, but they do not give a conceptual explanation for this fixed point.

1.2. What we do and do not prove. We use the following terminology throughout. Let $F(T, V)$ be the degree-23 equation of [1], with coefficients published in [2], and let $F_{\text{int}} \in \mathbf{Z}[T, V]$ be the equation obtained by clearing denominators as in [3]. By the *special fibre at $T = 0$* we mean the zero-dimensional scheme defined by $F_{\text{int}}(0, V)$; geometrically, it is the collection of roots of the specialized polynomial over $\overline{\mathbf{Q}}$, counted with multiplicity. Its seven simple roots are the unramified points of the cover over $T = 0$, and its eight double roots represent the ramification points. Here “double root” describes multiplicity in the fibre; it does not mean that the corresponding point is singular on the plane curve.

The investigation follows the factorization at $T = 0$ one step at a time.

- (1) The factor H_7 records the seven unramified points. We compute its Galois group and obtain a Fano plane on its roots. A resolvent gives an explicit polynomial for the seven lines (theorem 3.1 and proposition 3.2).
- (2) The factor R_8 records the eight ramification points. Adjoining its roots enlarges the splitting field of H_7 by a group $(\mathbf{Z}/2)^3$ (theorem 4.1).
- (3) Translation classes of ordered affine frames on these eight roots form a canonical 168-element $\text{PSL}_3(2)$ -torsor. Tame specialization identifies its arithmetic action with the action induced from the generic 23-sheet fibre (propositions 4.2 and 4.3).
- (4) The first nonzero term that separates the two local branches at a root of R_8 supplies one further square root. This produces a Galois extension with group $C_{M_{23}}(x) \cong 2^4 : \text{PSL}_3(2)$ (theorem 4.6). We then determine its ramification at 2, 3, and 23 (theorem 4.7).
- (5) An involution in $\text{PSL}_3(2)$ determines an incident point–line pair by its center and fixed line. This describes complex conjugation on the three real roots of H_7 , and it also applies when Frobenius at an unramified prime has cycle type $1^3 2^2$ (theorems 5.1 and 5.3).
- (6) Finally, we translate the complex-conjugation calculation back to the seven Nielsen classes and identify the three classes that it fixes (theorem 6.2).

These statements do not distinguish the Galois-fixed Nielsen class from the other six classes and do not imply its rationality. Their interest is more limited and concrete: they determine the Galois action and ramification visible in one singular fibre of the published cover. All new

finite-group and number-field assertions are verified by exact machine computation; §7 lists the certificates.

1.3. Relation to other work. The braid-orbit computation underlying $|\text{Ni}_c| = 7$ goes back to [4]. The numerical Belyi map algorithm used in [1] is that of Klug–Musty–Schiavone–Voight [5], implemented in the database of Musty–Schiavone–Sijssling–Voight [6].

Several ingredients are classical and are not claimed. The Gassmann phenomenon for $\text{PSL}_3(2)$ acting on points and lines is the standard example of arithmetically equivalent nonisomorphic fields. The correspondence between involutions of $\text{PSL}_3(2)$ and flags of the Fano plane is elementary projective geometry, and [theorem 5.1](#) combines it with ordinary decomposition theory; what is ours there is the identification of the archimedean flag with the real locus and the verification at primes whose Frobenius element is an involution.

In particular [theorem 4.6](#) is *not* a new solution of an existence problem for the abstract group $C_{M_{23}}(x)$. The exact sequence

$$1 \longrightarrow K \longrightarrow C_{M_{23}}(x) \longrightarrow \text{PSL}_3(2) \longrightarrow 1$$

is split, and embedding problems of this kind over Hilbertian fields are already known to be solvable [7]. The assertion here is that polynomials obtained from the fibre at $T = 0$ have the stated splitting fields and ramification.

2. THE LOCAL PICTURE AT THE BRANCH POINT $T = 0$

2.1. Group-theoretic notation. We fix once and for all a copy of $M_{23} \subset \text{Sym}(23)$ and elements

$$x \in 2A, \quad y \in 23A, \quad xy \in 23A,$$

so that $(x, y, (xy)^{-1})$ is a generating triple in Ni_c : indeed -1 is a nonresidue modulo 23, so $(xy)^{-1} \in 23B$. Write

$$C_{M_{23}}(x) = C_{M_{23}}(x), \quad K = O_2(C_{M_{23}}(x)), \quad H = \text{Fix}(x) \subset \{1, \dots, 23\}.$$

Here $O_2(C_{M_{23}}(x))$ denotes the largest normal subgroup of $C_{M_{23}}(x)$ whose order is a power of 2. Since x has cycle shape $1^7 2^8$, $|H| = 7$; the seven fixed letters form a *heptad*, that is, a seven-point block of the Witt design $S(4, 7, 23)$. One has

$$C_{M_{23}}(x) \cong 2^4 : \text{PSL}_3(2), \quad |C_{M_{23}}(x)| = 2688, \quad |K| = 16.$$

For reproducibility, the certificate labeling uses

$$\begin{aligned} x &= (3\ 21)(4\ 16)(9\ 22)(10\ 15)(11\ 20)(12\ 19)(13\ 18)(14\ 17), \\ y &= (1\ 2\ 11\ 10\ 16\ 9\ 6\ 3\ 23\ 19\ 20\ 14\ 21\ 17\ 4\ 8\ 22\ 5\ 18\ 15\ 13\ 7\ 12), \\ r &= (3\ 18)(4\ 16)(5\ 6)(7\ 23)(10\ 14)(12\ 19)(13\ 21)(15\ 17), \end{aligned}$$

The role of the explicit involution r will be explained in [theorem 6.2](#): it compares the published triple with its complex conjugate. These representatives generate the same published Nielsen class as the tuple of [1, §3]; all assertions below are invariant under simultaneous conjugation.

For fixed y let

$$X_y = \{x' \in 2A : x'y \in 23A\}, \quad |X_y| = 161 = 7 \cdot 23,$$

whose $\langle y \rangle$ -translation orbits are the seven Nielsen classes. We refer to the class containing x as the *published class*.

2.2. Branch points and the special fibre. We first recall only the parts of the branch cycle description needed below. They are due to Huang et al. The cover is branched over 0 and the two roots of $T^2 + 23$, with local monodromy $(2A, 23A, 23B)$ and total ramification at the order-23 points [1, §3]. The valuations at the two points above $T^2 + 23 = 0$, including the Newton-polygon edge from $(0, 0)$ to $(23, 4)$ that produces $\text{ord}_q(V) = -4$, are computed in [3, Lemma 2.1], and the singular scheme of the plane model is certified there as exactly 84 reduced *ordinary* nodes [3, Computation 4.1].

We record instead the corresponding factorization of the discriminant over $\mathbf{Z}$, which is used repeatedly below and which separates the three phenomena by explicit resultant tests rather than by inspection.

Proposition 2.1. *Let $F_{\text{int}} \in \mathbf{Z}[T, V]$ be the integral model, of degree 23 in V with leading coefficient $(T^2 + 23)^4$. Then*

$$\text{disc}_V(F_{\text{int}}) = T^8 (T^2 + 23)^{92} G(T)^2, \quad \deg G = 84,$$

where G is squarefree and divides $\gcd(\text{Res}_V(F, F_V), \text{Res}_V(F, F_T))$, while T does not. Consequently:

- (1) *the fibre over $T = 0$ consists of smooth points of the plane model, and $F_{\text{int}}(0, V) = 23^4 H_7 R_8^2$ gives cycle type $1^7 2^8$, the shape of class $2A$;*
- (2) *the 84 zeros of G are the nodes of [3, Computation 4.1], which being ordinary contribute no ramification to the normalization;*
- (3) *at the place $T^2 + 23$ the Newton polygon is a single segment of slope $4/23$; since $\gcd(4, 23) = 1$ the local extension is totally ramified of degree 23, recovering the 23-cycles at $T = \pm\sqrt{-23}$.*

Riemann–Hurwitz then gives

$$2g - 2 = -46 + 8 + 22 + 22 = 6, \quad g = 4.$$

The ramification point over $T^2 + 23$ lies at $V = \infty$ — equivalently $\text{ord}_q(V) = -4$ — which is what the leading coefficient $(T^2 + 23)^4$ encodes.

Remark 2.2. Comparing the two available plane models of the same curve isolates what is intrinsic from what is an artifact of the model. For the bidegree- $(23, 4)$ model P of [3] one has

$$\text{disc}_W P = T^8 (T^2 + 23)^{22} H_{62}(T)^2, \quad \deg H_{62} = 62,$$

up to a nonzero rational constant. The factor T^8 is the same in both: it is the genuine branch locus over the rational branch point, the eight ramification points represented by double roots in the $2A$ fibre. The square factors are not the same — 84 nodes against 62 — and neither is the exponent of $T^2 + 23$, both of which depend on the chosen projection. Only the T^8 and the presence of $T^2 + 23$ are intrinsic.

The following observation explains the quadratic field attached to the two nonrational branch points. It provides context for the fixed-point problem but is not used in the later calculations with H_7 and R_8 .

2.3. Why the cyclotomic obstruction vanishes.

Proposition 2.3. *y^λ is conjugate to y in M_{23} if and only if λ is a quadratic residue modulo 23; for the eleven nonsquares y^λ lies in the opposite class $23B$.*

Since the two order-23 branch points are the roots of $T^2 + 23$, conjugate over $\mathbf{Q}(\sqrt{-23})$ — the quadratic subfield of $\mathbf{Q}(\zeta_{23})$, fixed exactly by the squares — Proposition 2.3 identifies the branch swap with the nonsquare part of the cyclotomic character. This identifies the

square/nonsquare dichotomy with the action on the two nonrational branch points, and it explains the appearance of $\mathbf{Q}(\sqrt{-23})$ noted with an exclamation mark in [1, §3].

Remark 2.4. It also shows that the classical obstruction does not apply. Fried's branch cycle argument sends a cover with class tuple (C_1, C_2, C_3) under σ to one with $(C_1^\lambda, C_2^\lambda, C_3^\lambda)$, $\lambda = \chi(\sigma)$. Here $2A^\lambda = 2A$ always, while by [proposition 2.3](#) $23A^\lambda = 23A$ exactly for the eleven quadratic residues and $23B$ otherwise. So for nonsquare λ the tuple becomes $(2A, 23B, 23A)$ — the original with its last two entries exchanged — and the same σ exchanges the branch points $\pm\sqrt{-23}$, $\mathbf{Q}(\sqrt{-23})$ being the quadratic subfield of $\mathbf{Q}(\zeta_{23})$ fixed by the squares. The branch cycle argument therefore *permits* field of moduli $\mathbf{Q}$ without forcing it. That is the precise sense in which the phenomenon of [1, §1.4] is allowed but unexplained: the classical obstruction vanishes, and something finer decides.

3. THE FANO PLANE ON THE SEVEN UNRAMIFIED POINTS

3.1. The fibre over $T = 0$. Specializing at the rational branch point $T = 0$ gives

$$F_{\text{int}}(0, V) = 23^4 H_7(V) R_8(V)^2$$

with H_7 and R_8 irreducible over $\mathbf{Q}$ of degrees 7 and 8. The seven roots of H_7 are the unramified sheets fixed by the local $2A$ inertia; the eight roots of R_8 are the ramification points, represented by double roots of the specialized polynomial. Explicitly,

$$\begin{aligned} H_7(V) &= V^7 - 312V^6 + 43848V^5 - 3914896V^4 + 241123008V^3 \\ &\quad - 9876755712V^2 + 251799776256V - 2694891036672, \\ R_8(V) &= V^8 + 156V^7 - 1290V^6 - 1217744V^5 - 40611765V^4 \\ &\quad + 2044635024V^3 + 115083889782V^2 + 968032514052V - 6058197515007. \end{aligned}$$

Theorem 3.1. $\text{Gal}(H_7) \cong \text{PSL}_3(2)$, acting on the seven unramified sheets. The 21 involutions of that group each fix exactly three sheets. There are seven distinct fixed triples, each arising from three involutions. These seven triples form a $2-(7, 3, 1)$ design: every sheet lies in three triples, and each of the $\binom{7}{2} = 21$ pairs of sheets lies in exactly one. Thus the sheets are the points of a Fano plane and the seven fixed triples are its lines.

In what follows, a *flag* means an incident point–line pair in this Fano plane.

Proposition 3.2. Let $\alpha_1, \dots, \alpha_7$ be the roots of H_7 . The three-subset-sum resolvent is

$$\mathcal{R}_3(Y) = \prod_{1 \leq i < j < k \leq 7} (Y - (\alpha_i + \alpha_j + \alpha_k)).$$

It has degree 35 and factors over $\mathbf{Q}$ into irreducible factors of degrees 7 and 28. The degree-seven factor is

$$\begin{aligned} L_7(Y) &= Y^7 - 936Y^6 + 379728Y^5 - 86588640Y^4 + 11922182400Y^3 \\ &\quad - 982633068288Y^2 + 44572755885824Y - 859311381083136 \end{aligned}$$

and its seven roots are the sums of the three points on the seven Fano lines. The fields $\mathbf{Q}[V]/(H_7)$ and $\mathbf{Q}[Y]/(L_7)$ are nonisomorphic, but they have the same Galois closure and the same Dedekind zeta function. Their common field discriminant

$$2^4 3^6 23^6 = 1726690609296;$$

the point and line stabilizers are nonconjugate subgroups of order 24 with equal permutation characters. Thus these are the classical example of arithmetically equivalent but nonisomorphic fields arising from a Gassmann pair.

We now relate this Fano plane to the local inertia involution x inside M_{23} . An involution of $\mathrm{PSL}_3(2)$ acts on the Fano plane as a *transvection*: it fixes one line pointwise, called its axis, and has a distinguished fixed point on that line, called its center. The full correspondence between involutions and flags is proved in [theorem 5.1](#).

Proposition 3.3. *The points, lines, and flags have the following description using the action of $C_{M_{23}}(x) = C_{M_{23}}(x)$. All 99 involutions of $C_{M_{23}}(x)$ belong to the M_{23} conjugacy class $2A$, and:*

- (1) *the fifteen nonidentity elements of K are involutions and fix H pointwise. The central element x forms one conjugacy class in $C_{M_{23}}(x)$, and the other fourteen form a second. The map sending a noncentral $k \in K$ to the unique line invariant under $C_{C_{M_{23}}(x)}(k)$ identifies*

$$\{\text{lines}\} = (K/\langle x \rangle) \setminus \{0\};$$

writing $W = K/\langle x \rangle \cong \mathbf{F}_2^3$, the plane is thus $\mathbf{P}(W^)$ on the sheets and $\mathbf{P}(W)$ on the lines, which explains why the two degree-seven Galois sets are dual rather than isomorphic;*

- (2) *as an $\mathbf{F}_2[C_{M_{23}}(x)]$ -module, K is indecomposable. Its unique minimal nonzero submodule is the one-dimensional space $\langle x \rangle$, and the quotient $K/\langle x \rangle$ has dimension three. Consequently each nonzero element of W has exactly two preimages in K , exchanged by multiplication by x ;*
- (3) *the 84 involutions of $C_{M_{23}}(x) \setminus K$ form a single class and act on H as transvections. Sending such an involution to its center and fixed line gives a surjection onto the 21 flags, with four involutions above each flag. For such an involution t ,*

$$C_K(t) = \langle x \rangle \cup \{\text{the two lifts of the axis}\}, \quad \mathrm{Fix}(t) \cap H = \text{axis}(t).$$

Part (ii) says precisely that there is no $C_{M_{23}}(x)$ -invariant complement to $\langle x \rangle$ in K : the exact sequence

$$1 \longrightarrow \langle x \rangle \longrightarrow K \longrightarrow K/\langle x \rangle \longrightarrow 1$$

does not split as a sequence of $\mathbf{F}_2[C_{M_{23}}(x)]$ -modules.

3.2. Rigidity of the duality. The outer point–line duality of $\mathrm{PSL}_3(2)$ is not induced by an automorphism of $C_{M_{23}}(x)$ or by a permutation of the 23 sheets that normalizes $C_{M_{23}}(x)$:

$$\mathrm{Aut}(C_{M_{23}}(x)) = \mathrm{Inn}(C_{M_{23}}(x)) \quad (\text{order } 1344), \quad N_{\mathrm{Sym}(23)}(C_{M_{23}}(x)) = C_{M_{23}}(x),$$

and the point- and line-stabilizer preimages are nonconjugate of order 384. Hence no such automorphism or sheet permutation exchanges the two seven-element $C_{M_{23}}(x)$ -sets.

4. NUMBER FIELDS OBTAINED FROM THE FIBRE

We now follow the two factors of the special fibre through their splitting fields. Let

$$E = \mathrm{Spl}_{\mathbf{Q}}(H_7), \quad L_0 = \mathrm{Spl}_{\mathbf{Q}}(H_7 R_8).$$

This section constructs a third field M and proves

$$\mathbf{Q} \subset E \subset L_0 \subset M, \quad [E : \mathbf{Q}] = 168, \quad [L_0 : E] = 8, \quad [M : L_0] = 2.$$

The three Galois groups over $\mathbf{Q}$ are, successively, $\mathrm{PSL}_3(2)$, $2^3:\mathrm{PSL}_3(2)$, and $2^4:\mathrm{PSL}_3(2)$. Thus each step records additional information from the fibre: first the seven unramified points, then the eight ramification points, and finally the choice between the two local branches at each ramification point.

4.1. The splitting field of H_7R_8 . The local decomposition group at $T = 0$ centralizes the inertia involution x . The element x exchanges the two local branches at a ramification point but fixes the corresponding root of R_8 . Therefore the action on the eight roots of R_8 factors through $C_{M_{23}}(x)/\langle x \rangle$.

This quotient is the affine group of the three-dimensional vector space $W = K/\langle x \rangle \cong \mathbf{F}_2^3$. After identifying the eight roots of R_8 with the eight points of this affine space, the difference of two roots is a nonzero vector of W . The seven nonzero vectors correspond, by [proposition 3.3](#), to the seven Fano lines. Each nonzero difference occurs for four of the 28 unordered pairs of affine points.

Theorem 4.1. *The Galois group of the splitting field of H_7R_8 is $2^3:\mathrm{PSL}_3(2)$, of order 1344, acting affinely on the eight roots of R_8 . The restriction map $\mathrm{Gal}(L_0/\mathbf{Q}) \rightarrow \mathrm{Gal}(E/\mathbf{Q})$ has kernel $K/\langle x \rangle \cong (\mathbf{Z}/2)^3$.*

Proof sketch. Decomposition theory at the tame rational place $T = 0$, together with the equality of arithmetic and geometric monodromy M_{23} [1], embeds $\mathrm{Gal}(L_0/\mathbf{Q})$ in $N_{M_{23}}(\langle x \rangle)/\langle x \rangle = C_{M_{23}}(x)/\langle x \rangle$. Restriction to E is surjective. Its kernel is therefore a submodule of the irreducible three-dimensional module $K/\langle x \rangle$, so it is either trivial or the whole module. At $p = 599$, H_7 and L_7 split completely while R_8 factors into four irreducible quadratics. Thus Frobenius acts trivially on E but as a nonzero translation on the roots of R_8 , proving that the kernel is the whole module. $\square$

Independently, a direct number-field computation gives $[\mathrm{Spl}_{\mathbf{Q}}(R_8) : \mathbf{Q}] = 1344$.

4.2. Affine frames and specialization. Let A be the eight-element set of roots of R_8 and put $W = K/\langle x \rangle \cong \mathbf{F}_2^3$. The group W acts regularly on A . An *ordered affine frame* is a tuple (a_0, a_1, a_2, a_3) for which the three differences $a_i - a_0$, $1 \leq i \leq 3$, form a basis of W . Let

$$\mathcal{F}(A) = \{\text{ordered affine frames of } A\}/W,$$

where W acts by simultaneous translation.

Proposition 4.2. *The set $\mathcal{F}(A)$ has 168 elements. The action of $C_{M_{23}}(x)/\langle x \rangle$ on affine frames descends to a free transitive action of*

$$C_{M_{23}}(x)/K \cong \mathrm{PSL}_3(2)$$

on $\mathcal{F}(A)$. Via the difference labelling of [proposition 3.3](#), a point of $\mathcal{F}(A)$ is equivalently an ordered basis of the Fano line module.

Proof. There are eight choices of origin. Once the origin is chosen, there are 7, 6, and 4 choices for the successive basis vectors, so there are $8 \cdot 7 \cdot 6 \cdot 4 = 1344$ affine frames. Translation acts freely on the origins; hence the quotient has $1344/8 = 168$ elements. The translation subgroup W is exactly the kernel on translation classes, and the linear quotient $\mathrm{PSL}_3(2) = \mathrm{GL}_3(2)$ acts freely and transitively on ordered bases of W . The seven nonzero vectors of W are the seven canonically labelled Fano lines. $\square$

We next record a formal comparison which will be used in the companion paper. Let S be the marked geometric generic fibre, based at the rational tangential basepoint at $T = 0$ defined by the formal direction $T \rightarrow 0$. This choice fixes the fibre functor and the orientation used to record branch cycles. For $\sigma \in \mathfrak{G}_{\mathbf{Q}}$, let $d(\sigma)$ be the descent permutation of S . The branch-cycle lemma sends tame inertia to its $\chi(\sigma)$ -th power. Since x has order two, $x^{\chi(\sigma)} = x$, and the equality of arithmetic and geometric monodromy places $d(\sigma)$ in $C_{M_{23}}(x)$. Write $\rho_8 : \mathfrak{G}_{\mathbf{Q}} \rightarrow \mathrm{Sym}(A)$ for the ordinary Galois action on the roots of R_8 , and let $\mathrm{pair}(c)$ be the permutation induced

by $c \in C_{M_{23}}(x)$ on the eight two-element $\langle x \rangle$ -orbits, transported to A by the specialization bijection in the proof below.

Proposition 4.3 (Generic–special-fibre compatibility). *For every $\sigma \in \mathfrak{S}_{\mathbf{Q}}$,*

$$\rho_8(\sigma) = \text{pair}(d(\sigma)).$$

Consequently the induced Galois action on $\mathcal{F}(A)$ is the action obtained from the generic-fibre descent permutation.

Proof. Let $R = \mathbf{Q}[T]_{(T)}^h$ and normalize $\text{Spec } R$ in the cover. The scheme $\text{Spec } R$ is a henselian trait, meaning the spectrum of a henselian discrete valuation ring; its punctured trait is the complement of its closed point. At a geometric point of ramification index e , tameness and smoothness of the total curve give the completed local form $T = u^e$ after extracting the e -th root of a unit. The e sheets over the punctured trait form one inertia orbit and specialize to one geometric point. Hence specialization gives a canonical bijection

$$S/\langle x \rangle \xrightarrow{\sim} \{\text{geometric points above } T = 0\}.$$

Normalization and specialization commute with base-field automorphisms, so this bijection is $\mathfrak{S}_{\mathbf{Q}}$ -equivariant. Restricting it to the eight two-element inertia orbits identifies them with the eight roots of R_8 and gives the displayed equality. Passing functorially to affine frames and then quotienting by translations proves the final assertion. This is the finite-set case of tame specialization; see [8, Exposé XIII, §2.10]. $\square$

Remark 4.4. The companion paper [9] identifies the descent permutation in [proposition 4.3](#) with the homomorphic Grothendieck–Teichmüller frame return and constructs a marked point–line correlation through $\mathcal{F}(A)$. That is a compatibility theorem for the explicit $\mathbf{Q}$ -cover; it does not supply a preferred rational affine frame and does not reprove rationality without the known equation.

4.3. Separating the two branches: the final quadratic extension.

Proposition 4.5. *$C_{M_{23}}(x)$ is perfect, its center is $\langle x \rangle$, and the central extension*

$$1 \longrightarrow \langle x \rangle \longrightarrow C_{M_{23}}(x) \longrightarrow C_{M_{23}}(x)/\langle x \rangle \longrightarrow 1$$

does not split. The group acts faithfully and transitively on the sixteen local branches above the eight roots of R_8 ; a branch stabilizer is isomorphic to $\text{PSL}_3(2)$.

The polynomial R_8 records the ramification points but does not distinguish the two local branches above each one. Near a root v of R_8 , the two branches have expansions

$$V = v \pm \sqrt{e(v)T} + O(T), \quad e(v) = -\frac{2F_T(0, v)}{F_{VV}(0, v)},$$

where subscripts denote partial derivatives. Since $F_{VV}(0, v) = 2 \cdot 23^4 H_7(v) R_8'(v)^2$, the square class of $e(v)$ is represented by the value at v of the polynomial

$$c(V) \equiv -F_T(0, V) H_7(V) \pmod{R_8}.$$

Theorem 4.6. *The eight values $c(v)$, for v a root of R_8 , represent one square class in $L_0^\times/L_0^{\times 2}$, and that class is nontrivial. If $M = L_0(\sqrt{c(v)})$ for any root v , then*

$$\text{Gal}(M/\mathbf{Q}) \cong C_{M_{23}}(x) = 2^4 : \text{PSL}_3(2)$$

and $[M : L_0] = 2$. Under this isomorphism, the nontrivial element of $\text{Gal}(M/L_0)$ is the inertia element x .

Proof sketch. Tame local theory identifies the splitting field over $\mathbf{Q}((T))$ as

$$\mathcal{L} = L_0((\sqrt{cT})).$$

Its Galois group is $C_{M_{23}}(x)$. Because $C_{M_{23}}(x)$ is perfect, $\mathcal{L}$ has no nontrivial abelian subextension of $\mathbf{Q}((T))$; in particular $\sqrt{T} \notin \mathcal{L}$, and c is not a square in L_0 . After adjoining $\sqrt{T}$ one has

$$\text{Gal}(\mathcal{L}(\sqrt{T})/\mathbf{Q}((T))) \cong C_{M_{23}}(x) \times \mathbf{Z}/2.$$

The diagonal involution $(x, -1)$ fixes $\sqrt{c} = \sqrt{cT}/\sqrt{T}$. Its fixed field is $M((T))$, where $M = L_0(\sqrt{c})$; this follows as well by comparing degrees. Taking the constant fields therefore gives

$$\text{Gal}(M/\mathbf{Q}) \cong (C_{M_{23}}(x) \times \mathbf{Z}/2)/\langle (x, -1) \rangle \cong C_{M_{23}}(x).$$

The exact arithmetic certificates give additional checks. The polynomial $X^2 - dc(V)$ is irreducible over $\mathbf{Q}[V]/(R_8)$ for

$$d \in \{\pm 1, \pm 2, \pm 3, \pm 6, \pm 23, \pm 46, \pm 69, \pm 138\}.$$

At each of the fourteen primes below 3×10^5 that split completely in L_0 , the eight values $c(v_i)$ have the same quadratic character. At six of those primes, beginning with $p = 13049$, that common character is -1 , which is impossible if c is a square in L_0 . $\square$

The polynomial R_8 has four real roots. For exactly two of them one has $c(v) > 0$, so both local branches are real; for the other two, $c(v) < 0$ and the branches are complex conjugate. This agrees with the action of r , which fixes the branches above two real ramification points and exchanges the branches above the other two.

4.4. Ramification.

Theorem 4.7. *Let $E_8 = \mathbf{Q}[V]/(R_8)$, let v be the image of V in E_8 , and put $E'_8 = E_8(\sqrt{c(v)})$. The number fields $\mathbf{Q}[V]/(H_7)$, $\mathbf{Q}[Y]/(L_7)$, and E_8 have discriminants $2^4 3^6 23^6$, $2^4 3^6 23^6$, and $2^{16} 3^6 23^6$, respectively. They are therefore unramified outside $\{2, 3, 23\}$. The relative discriminant of E'_8/E_8 has norm $2^8 3^2 23^2$. Consequently its Galois closure M is unramified outside the same three rational primes, and the central quadratic extension M/L_0 ramifies above each of them. More precisely:*

- (1) *at $p = 23$, both degree-seven fields are totally ramified with $e = 7$ and all residue degrees 1. In L_0 , inertia is cyclic of order 7 and acts transitively on the seven Fano points and on the seven Fano lines; such an element is called a Singer cycle. It fixes exactly one root of R_8 , the affine origin, and permutes the other seven cyclically, giving the octic factorization pattern $1 + 7$. These factorization patterns do not by themselves determine the decomposition group: tame Frobenius acts by $\tau \mapsto \tau^{23} = \tau^2$, of order three on C_7 , so the decomposition group is 7:3. Both octic primes above 23 ramify in M/L_0 ; in the full degree-2688 field inertia is C_{14} and decomposition is $C_2 \times (7:3)$;*
- (2) *at $p = 3$ the ramification is wild with $e = 3$ and lies in the image $\text{PSL}_3(2)$ of $C_{M_{23}}(x)/K$;*
- (3) *at $p = 2$ the degree-seven fields are tame with $e = 3$ while the degree-eight field is wild with $e = 4$. Since the ramification indices in the degree-seven fields are odd, the wild 2-group visible in the affine quotient acts trivially on the seven Fano points and lies in $K/\langle x \rangle$. Among the six classes of order-12 subgroups of $C_{M_{23}}(x)$ exactly one has the required orbit lengths and has its Sylow 2-subgroup inside K ; it is an A_4 , with Klein four wild part and tame quotient C_3 .*

Proof sketch. The three absolute discriminants and the displayed (e, f) data are computed from maximal orders. For E'_8/E_8 , the exact polynomial $c(V)$ of [theorem 4.6](#) gives $\mathfrak{d}_{E'_8/E_8}$ by a relative discriminant computation; its ideal norm is $2^8 3^2 23^2$, with both primes of E_8 above 23

appearing to exponent one. At 23, tame local theory gives $\phi\tau\phi^{-1} = \tau^{23} = \tau^2$; since 2 has order three modulo 7, the residue decomposition group is 7:3. The quadratic extension adjoins x to inertia, giving C_{14} , and its normalizer in $C_{M_{23}}(x)$ has order 42 and quotient C_3 . The assertions at 2 and 3 follow from the displayed factorization and ramification data for H_7 , L_7 , and R_8 , together with the exact subgroup census recorded in the certificates. $\square$

5. FLAGS SELECTED BY COMPLEX CONJUGATION AND FROBENIUS

We now use decomposition groups of the field E to select flags of the Fano plane. Recall that a flag is an incident point–line pair. Although $\text{Gal}(E/\mathbf{Q}) \cong \text{PSL}_3(2)$ fixes no individual point, line, or flag, the decomposition group of a chosen place may do so.

Theorem 5.1. *Every involution of $\text{PSL}_3(2)$ acts on the Fano plane as a transvection. It fixes one line pointwise, called its axis, and has a distinguished fixed point on that line, called its center. The map*

$$t \longmapsto (\text{center}(t), \text{axis}(t))$$

is a bijection from the 21 involutions to the 21 flags.

On the seven points, t has cycle type $1^3 2^2$: its three fixed points are the points of its axis, and the remaining four points form two transposed pairs. Besides the axis, exactly two lines are invariant under t ; each consists of the center together with one transposed pair. The group is transitive on flags, each flag has stabilizer of order 8, and each involution has centralizer of order 8.

The field E has no real embeddings and has 84 conjugate pairs of complex embeddings. The decomposition group of each complex place is generated by an involution t , so the place determines the flag $(\text{center}(t), \text{axis}(t))$. In this way the 84 complex places map four-to-one onto the 21 flags. The same construction applies to a finite place of E above an unramified rational prime whenever its Frobenius element is an involution.

Corollary 5.2. *For complex conjugation, the axis is the set of three real roots of H_7 . Their sum is a root of L_7 , so they form a Fano line. The center is the unique real root that lies on each of the other two invariant lines.*

More explicitly, the four nonreal roots form two conjugate pairs. For each real root, count how many of those pairs combine with it to form a Fano line. The three counts are $(0, 0, 2)$; the count 2 occurs at the largest real root, approximately 102.2684.

This real Fano line is defined over $\mathbb{R}$ but not over $\mathbf{Q}$. Indeed, $\text{Gal}(E/\mathbf{Q}) \cong \text{PSL}_3(2)$ acts transitively on the seven lines and on the 21 flags.

The same structure occurs at finite places. Suppose that p is unramified and that H_7 has factorization type $1^3 2^2$ modulo p . Then Frobenius is an involution. By Chebotarev, such primes have density $21/168 = 1/8$.

Theorem 5.3. *Let N_{28} be the degree-28 factor of $\mathcal{R}_3$ whose roots are the sums of triples that are not Fano lines, and exclude the finitely many primes dividing $\text{Res}(L_7, N_{28})$. At every remaining unramified prime for which H_7 has factorization type $1^3 2^2$, the three roots in $\mathbf{F}_p$ sum to a root of L_7 modulo p and hence form the axis.*

The other four roots form two Frobenius-conjugate pairs. For each of the three roots in $\mathbf{F}_p$, count how many of these pairs combine with it to form a Fano line. The three counts are $(0, 0, 2)$, and the root with count 2 is the center. The group-theoretic argument proves this for every prime satisfying the hypotheses; the exact computation independently checks all 9,889 such primes below 10^6 without exception.

The resultant exclusion is necessary. Below 10^6 there is exactly one collision between the two resolvent factors, at $p = 22159$, and 22159 divides $\text{Res}(L_7, N_{28})$.

The real place and the prime 23 give complementary pictures of the same Fano plane. Complex conjugation fixes one line. At 23 , a generator of tame inertia cyclically permutes all seven points and all seven lines. After numbering the points cyclically modulo 7 , one line is a three-element subset whose nonzero differences occur exactly once, and the other six lines are its translates.

6. COMPLEX CONJUGATION ON THE NIELSEN CLASSES

6.1. Representing all seven classes with the same involution. Complex conjugation sends a triple (x', y', z') to (x', z'^{-1}, y'^{-1}) . To compare the resulting triples, choose the representative in which the first element is the fixed involution x and let

$$Y_x = \{y'' \in 23A : xy'' \in 23A\},$$

so that pairs (x, y'') with $y'' \in Y_x$ represent the seven classes. The centralizer $C_{M_{23}}(x)$ acts on Y_x by conjugation.

Proposition 6.1. *Call a pair (x', y') admissible if $x' \in 2A$, $y' \in 23A$, and $x'y' \in 23A$. The group M_{23} acts freely on the admissible pairs by simultaneous conjugation. Moreover,*

$$|Y_x| = 18816 = 7 \times 2688,$$

and Y_x is the disjoint union of seven free $C_{M_{23}}(x)$ -orbits, one for each Nielsen class. Equivalently,

$$|Y_x| = |X_y| \frac{|23A|}{|2A|} = 161 \cdot \frac{443520}{3795} = 18816.$$

For the triple $(x, y'', (xy'')^{-1})$, complex conjugation exchanges the two order-23 entries and inverts them. After writing the result again with first entry x , the second entry is xy'' . Thus, in these coordinates, complex conjugation is the permutation

$$\mu : Y_x \longrightarrow Y_x, \quad \mu(y'') = xy''.$$

This is the point at which the investigation returns from the arithmetic of the special fibre to the original seven Nielsen classes. To decide whether a class is fixed, we ask whether an element of $C_{M_{23}}(x)$ conjugates y'' to $\mu(y'')$.

Theorem 6.2. *For a representative (x, y'') , the equation*

$$y''^w = xy'', \quad w \in C_{M_{23}}(x),$$

has a solution if and only if the corresponding Nielsen class is fixed by complex conjugation. We call such a solution w the comparison element. Exactly three classes are fixed. For each of them the comparison element is unique; it is an involution in $C_{M_{23}}(x) \setminus K$ and hence induces a transvection on the seven roots of H_7 . For the published class, $w = r$: this is the unique element of M_{23} satisfying $x^r = x$ and $y^r = xy$. The center and axis of the transvection induced by r are the flag associated with the real place in [theorem 5.1](#).

Thus the comparison for the published class is the explicit involution $r \in C_{M_{23}}(x)$; its order-two condition is simply $r^2 = 1$. Since r acts on the Fano plane as a collineation, it does not interchange the point and line sets.

The real place confirms the picture arithmetically: r acts on the seven roots of H_7 with cycle type $1^3 2^2$, fixes three Fano lines, and fixes four of the eight roots of R_8 . Correspondingly, H_7 , L_7 , and R_8 have exactly 3, 3, and 4 real roots.

The results of this section describe complex conjugation only. They do not determine the action of the full absolute Galois group on the seven Nielsen classes and therefore do not explain why the published class is fixed.

7. CERTIFICATES

Every new finite and local computation above is verified by an exact machine certificate; none relies on floating-point heuristics except where a numerical identification is accompanied by an explicit separation guard. The assertion that the 84 singularities are ordinary nodes is imported from the separately cited optimal-model singular-scheme computation; the in-repository resultant certificate locates the singular fibres but does not reprove ordinarity. The table lists the reproduction targets in the accompanying repository <https://github.com/research-reflexivity/frontier-math-m23-fixed-point>.

Result	Target	Engine
Prop 2.1, Prop 2.3	branch-cycle-description	GAP+PARI
Thm 3.1, Props 3.2, 3.3	special-fibre-fano, fano-flag-descent	GAP+PARI
Thm 5.1, 5.3	local-flag-places	GAP+PARI
Thm 4.1	special-fibre-field-tower, octic-splitting-field	GAP+PARI
Thm 4.6	vanishing-cycle-class	GAP+PARI
Thm 4.7	special-fibre-ramification	GAP+PARI
Thm 6.2	real-frame-cocycle	GAP+PARI

The largest verifications are the direct computation of the degree-1344 splitting field of R_8 , the check of the ten possible pairs of factorization types at 348,508 Frobenius elements, the incidence check at the 9,889 primes below 10^6 for which H_7 has factorization type $1^3 2^2$, and the square-class check at the fourteen tested primes that split completely in L_0 .

8. OPEN QUESTIONS

The computations leave the following concrete questions.

- (1) Compute equations for the other six covers in the Nielsen class and thereby determine the full Galois action on the seven covers. Complex conjugation acts on the six non-published classes with cycle type $1^2 2^2$, so their degree-six algebra has signature $(2, 2)$.
- (2) Classify the 2^4 : $\mathrm{PSL}_3(2)$ -extensions of $\mathbf{Q}$ unramified outside $\{2, 3, 23\}$ with the local groups and constituent fields of [theorem 4.7](#). In particular, determine whether the field M constructed in [theorem 4.6](#) is unique with these properties.
- (3) Determine whether the quadratic field governing the two rulings of the canonical quadric in [\[1, Remark 3.4\]](#) is $\mathbf{Q}(\sqrt{-23})$, the quadratic field that exchanges the two order-23 branch points here.
- (4) Decide whether any invariant of the fibre at $T = 0$ is functorial under the absolute Galois action and distinguishes the published Nielsen class without first using its rational equation. The affine-frame compatibility of [proposition 4.3](#) is functorial for the published cover but does not distinguish it among the seven classes.

9. SCOPE, PROVENANCE, AND DISCLOSURE

The M_{23} theorem, the degree-23 equation, the branch cycle description, and the observation that the Galois action on the seven Nielsen classes has a fixed point are due to Huang et al. [\[1\]](#). The integral model used throughout, the valuations at the points over $T^2 + 23 = 0$, and

the certification of the plane model's singular scheme as 84 ordinary nodes are from [3]. The Gassmann phenomenon for $\mathrm{PSL}_3(2)$ on points and lines is classical.

The contribution here is the factorization of the fibre at $T = 0$ and the following deductions from it: the explicit point and line fields; the description of their Fano-plane incidence in terms of $C_{M_{23}}(x)$; the local description at the real place and at primes whose Frobenius is an involution; the construction of the degree-1344 and degree-2688 Galois extensions; their ramification groups; the affine-frame torsor and its generic-special-fibre comparison; and the explicit comparison element for each class fixed by complex conjugation.

We do not prove that the fixed point is rational; that is a theorem of [1], established there by explicit computation. Nor do we give a new criterion that forces a fixed point in this Nielsen class. The six conjugate covers are not computed here, and no claim is made about the action of general elements of $\mathfrak{S}_{\mathbf{Q}}$ beyond the decomposition-group and complex-conjugation statements proved above.

The discovery and drafting process used substantial AI assistance. The repository contains a full disclosure.

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